A. Truth Tables and Connectives

Core vocabulary

A **proposition** has a truth value. A **valuation** assigns T or F to each atomic proposition. A **tautology** is true on every row, a **contradiction** on none, and a **contingency** on some but not all. Two expressions are **logically equivalent** when their final columns match on every valuation.

Connective meanings

p	q	$\neg p$	$p \wedge q$	$p \vee q$
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

$p \to q$ is false only when p is T and q is F. $p \leftrightarrow q$ is true exactly when p and q match. "p only if q" and "q is necessary for p" both mean $p \to q$.

Building an eight-row table

For atoms p, q, r, list rows in a fixed binary pattern:

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Add one column for each useful subexpression, working from the innermost connectives outward. Classify only the final column.

A worked example

Classify $(p \to q) \vee (\neg p \to r)$.

1. Make the eight rows for p, q, r.
2. Compute $p \to q$ and $\neg p \to r$ in separate columns.
3. Join those columns with $\vee$.

As a check,

$$(p \to q) \vee (\neg p \to r) \equiv (\neg p \vee q) \vee (p \vee r).$$

Reorder the disjuncts to expose $\neg p \vee p$, which is always T. Therefore the final column must be all T, so the proposition is a **tautology**. On an assessment, the equivalence checks the complete table; it does not replace it.

Interpreting a row

Evaluate $(p \to q) \wedge (\neg q \vee r)$ when $p = F, q = T, r = F$:

$$\begin{aligned} p \to q &= F \to T = T, \\ \neg q \vee r &= F \vee F = F, \\ (p \to q) \wedge (\neg q \vee r) &= T \wedge F = F. \end{aligned}$$

Compute the inner pieces first and the main connective last. That dependency order should match the table's columns.

Common errors

Do not read $p \to q$ as a causal claim or as $q \to p$. Do not omit rows, change row order midway, or classify from a few sample rows. In $\neg(p \wedge q)$ the negation applies to the entire parenthesized expression.

A readiness check

Can you make all eight rows; compute $\neg, \wedge, \vee, \to, \leftrightarrow$; show intermediate columns; and classify the final proposition correctly?

B–C. Translation, Quantifiers, and Negation

Conditionals in ordinary language

"p if q"	$q \to p$
"p only if q"	$p \to q$
"p iff q"	$p \leftrightarrow q$
"p unless q"	$\neg q \to p$, equivalently $p \vee q$
"p whenever q"	$q \to p$

The antecedent is sufficient; the consequent is necessary.

B: conditional translation

Let e mean "the account may edit" and a mean "the account is an administrator." "The account may edit only if it is an administrator" is $e \to a$. Here editing is the claim and administrator status is necessary. Check the direction by reading it back: if editing occurs, administrator status must hold. Writing $a \to e$ instead says every administrator may edit.

Predicates and quantifiers

A **predicate** becomes a proposition after its variables are assigned or quantified. State the domain. Then

$$\forall x\, P(x): \ P \text{ holds for every } x,$$
$$\exists x\, P(x): \ P \text{ holds for at least one } x.$$

The order matters: $\exists x \forall y\, R(x,y)$ uses one x for every y; $\forall y \exists x\, R(x,y)$ may use a different x for each y.

B: nested quantifiers and argument order

Let $M(x,y)$ mean "student x messaged student y." Then

$$\exists x \quad \underbrace{\forall y\, M(x,y)}_{\text{same } x \text{ messages every } y}$$

says one student messaged every student. By contrast, $\forall y \exists x\, M(x,y)$ allows a different sender x for each recipient y. Finally, replacing $M(x,y)$ by $M(y,x)$ reverses sender and recipient. Check both the quantifier order and the predicate's argument order.

Restricting an all-things domain

If the domain is all things, use predicates to restrict types:

$$\text{Every } A \text{ is } B: \forall x(A(x) \to B(x)),$$
$$\text{Some } A \text{ is } B: \exists x(A(x) \wedge B(x)),$$
$$\text{No } A \text{ is } B: \forall x(A(x) \to \neg B(x)).$$

Use implication with a universal restriction and conjunction with an existential restriction.

C: push a negation inward

$$\neg \forall x\, P \equiv \exists x\, \neg P, \qquad \neg \exists x\, P \equiv \forall x\, \neg P.$$

Work from left to right: (1) flip the first quantifier crossed by the negation, (2) keep the remaining quantified formula in the same order, and (3) repeat. Then use De Morgan's laws and conditional elimination until each negation is immediately before a predicate.

C: a complete negation

Domain: all things. Let $D(x)$ mean "x is a dog" and $F(x)$ mean "x has fleas."

$$\text{Every dog has fleas}: \ \forall x(D(x) \to F(x)),$$
$$\text{negate } \forall: \ \exists x \neg(D(x) \to F(x)),$$
$$\text{negate the implication}: \ \exists x(D(x) \wedge \neg F(x)).$$

The final formula gives a counterexample to the original. English: some dog does not have fleas.

C: nested negation

$$\neg \forall x \exists y\, P(x,y) \equiv \exists x \neg \exists y\, P(x,y)$$
$$\equiv \exists x \forall y \neg P(x,y).$$

The first step flips $\forall x$ to $\exists x$; the second flips $\exists y$ to $\forall y$. The variables stay in their original order.

Common errors

Do not swap $\forall$ and $\exists$ without moving a negation through them. Do not reuse a single existential witness unless the formula requires one. Define both predicate meanings and argument order.

B–C readiness

Can you translate "if," "only if," and "unless"; distinguish nested quantifier orders; diagnose reversed arguments; symbolize, negate, simplify, and translate back to English?

D. Consistency and Satisfiability

What the question asks

A collection of formulas is **consistent** or **satisfiable** when at least one truth assignment makes all formulas true. It is **inconsistent** when no such assignment exists. A satisfying assignment is a witness; a contradiction is a proof that no witness exists.

A reliable workflow

1. Define one atom for each simple statement.
2. Translate every sentence before reasoning.
3. Start with facts and implications whose antecedents are forced.
4. Propagate consequences using valid inference rules.
5. If no contradiction appears, finish a complete assignment and check every condition.
6. If a contradiction appears, cite a small subset that forces both p and $\neg p$.

Useful inference rules

$p,\ p \to q \therefore q$	Modus Ponens	(MP)
$\neg q,\ p \to q \therefore \neg p$	Modus Tollens	(MT)
$p \to q,\ q \to r \therefore p \to r$	Hypothetical Syllogism	
$p \lor q,\ \neg p \therefore q$	Disjunctive Syllogism	

From $p \leftrightarrow q$, use both $p \to q$ and $q \to p$.

D: prove inconsistency

Let m = multiuser, n = normal, k = kernel works, i = interrupt. Suppose

$$m \leftrightarrow n,\ n \to k,\ \neg k \lor i,\ \neg m \to i,\ \neg i.$$

1. $\neg k$ follows from $\neg k \lor i$ and $\neg i$ (DS).
2. $\neg n$ follows from $n \to k$ and $\neg k$ (MT).
3. $\neg m$ follows from $m \to n$ and $\neg n$ (MT).
4. i follows from $\neg m \to i$ and $\neg m$ (MP).

Now i contradicts the given $\neg i$, so the collection is inconsistent.

D: prove consistency by a witness

Let u = software upgraded, a = users have access, s = users can save. Translate:

$$u \to \neg a, \qquad a \to s, \qquad \neg s \to \neg u.$$

Try $(u, a, s) = (T, F, T)$:

$$T \to T = T, \qquad F \to T = T, \qquad F \to F = T.$$

All three formulas are true, so this checked witness proves consistency. To list all models, test all eight rows or give a complete case split.

D: find a minimal contradiction

Suppose $p \to q$, $q \to r$, p, and $\neg r$. From p and $p \to q$, infer q; from q and $q \to r$, infer r, contradicting $\neg r$. All four are needed for this chain, so they form a minimal inconsistent subset. A larger list that contains them proves inconsistency but does not answer a "minimal collection" request.

Listing all models

For three atoms, a truth table has eight rows. Translate each condition into a column, conjoin the condition columns, and keep exactly the rows where the conjunction is T. When reasoning by cases, state why each branch is complete.

Invalid shortcuts

Affirming the consequent ($p \to q, q \therefore p$) and denying the antecedent ($p \to q, \neg p \therefore \neg q$) are invalid. "Only if" points toward the necessary condition. A conditional with false antecedent is true.

D readiness

Can you define atoms, translate every condition, give a checked satisfying assignment or a short contradiction, list all models when asked, identify forced values, and isolate a minimal contradictory subset?

E. Equivalence Derivations with Cited Rules

Permitted rules

Each may be used in either direction:

$a \to b \equiv \neg a \vee b$	Implication	(Imp)
$\neg\neg a \equiv a$	Double Negation	(DN)
$\neg(a \vee b) \equiv \neg a \wedge \neg b$	De Morgan	(DM)
$\neg(a \wedge b) \equiv \neg a \vee \neg b$	De Morgan	(DM)
$a \vee (b \wedge c) \equiv (a \vee b) \wedge (a \vee c)$	Distributive	(Dist)
$a \wedge (b \vee c) \equiv (a \wedge b) \vee (a \wedge c)$	Distributive	(Dist)
$(a \circ b) \circ c \equiv a \circ (b \circ c)$	Associative	(Assoc)
$a \circ b \equiv b \circ a$	Commutative	(Comm)
$a \circ a \equiv a$	Idempotent	(Idem)

Here $\circ$ is either $\wedge$ or $\vee$ where appropriate. Reparenthesizing requires Associative; reordering requires Commutative.

How to plan the chain

Eliminate implications first. Compare the connective shape of the two sides. Use Distributive to factor or expand, De Morgan plus Double Negation to move a negation, and then clean up order or duplicates. Change only one justified feature per line and cite exactly one rule.

E worked example: factor a common term

Show $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$.

$(p \to q) \wedge (p \to r) \equiv (\neg p \vee q) \wedge (p \to r)$	Implication
$\equiv (\neg p \vee q) \wedge (\neg p \vee r)$	Implication
$\equiv \neg p \vee (q \wedge r)$	Distributive
$\equiv p \to (q \wedge r)$	Implication.

The common disjunct $\neg p$ is what makes the Distributive step possible.

E worked example: De Morgan structure

Show $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$.

$(p \to r) \wedge (q \to r) \equiv (\neg p \vee r) \wedge (q \to r)$	Implication
$\equiv (\neg p \vee r) \wedge (\neg q \vee r)$	Implication
$\equiv (r \vee \neg p) \wedge (\neg q \vee r)$	Commutative
$\equiv (r \vee \neg p) \wedge (r \vee \neg q)$	Commutative
$\equiv r \vee (\neg p \wedge \neg q)$	Distributive
$\equiv (\neg p \wedge \neg q) \vee r$	Commutative
$\equiv \neg(p \vee q) \vee r$	De Morgan
$\equiv (p \vee q) \to r$	Implication.

The two Commutative steps place the common r where Distributive can factor it.

E worked example: nested implication

$\neg p \to (q \to r) \equiv \neg\neg p \vee (q \to r)$	Implication
$\equiv p \vee (q \to r)$	Double Negation
$\equiv p \vee (\neg q \vee r)$	Implication
$\equiv (p \vee \neg q) \vee r$	Associative
$\equiv (\neg q \vee p) \vee r$	Commutative
$\equiv \neg q \vee (p \vee r)$	Associative
$\equiv q \to (p \vee r)$	Implication.

Each line changes exactly one feature and cites exactly one permitted rule.

Common errors

Do not cite "algebra" or "simplify." Do not silently move parentheses or terms. Do not use a truth table when a rule-by-rule derivation is requested. Make sure every intermediate expression is equivalent to the preceding one.

Whole-Core readiness

Can you complete a truth table; translate conditional and quantified language; negate nested quantifiers; prove consistency or inconsistency; and write an equivalence chain with one named rule per step? A live form selects one problem from each section A–E and is designed for 20 minutes.